

# MATSim in Production: Live Demand Prediction, Automated DRT Scenario Build and Visualisation for Non-Expert Users

Tom-Erik Kuhlen – MOIA GmbH, Hamburg, Germany – [tom-erik.kuhlen@moia.io](mailto:tom-erik.kuhlen@moia.io) – +49 151 65586031

## Motivation

Cities, public transport operators and consultancies increasingly have to decide how large an autonomous on-demand fleet needs to be, how to shape its service area, and whether the business case holds. Classical macroscopic models cannot answer this, because pooling, detours and dispatching emerge from microscopic interactions between individual requests and individual vehicles. MATSim can (Horni et al., 2016), and its demand-responsive transport (DRT) extension (Bischoff et al., 2016) represents such fleets directly, which is the configuration our platform runs. It is, however, accessible only to those able to work with configuration XML, build networks, write Java and provide compute infrastructure. The people who need the answers are rarely the people who can operate the tool.

This paper asks: how do you embed MATSim in a product such that non-experts obtain defensible results without learning MATSim? We present the decisions we took building a commercial platform around it, the reasoning behind them, and what we still consider unresolved.

## The MIA user flow

MIA, the Mobility Impact Analyzer, is a browser-based platform developed by Volkswagen and MOIA, a Volkswagen Group company that operates a ridepooling service in Hamburg and is rolling out autonomous ridepooling services in several cities worldwide. MIA lets its users assess new autonomous on-demand services; it structures the assessment as a four-step workflow: region exploration, demand estimation, simulation and analysis. The steps are worked through in sequence, and each one produces the input for the next. The platform is in production use with more than 200 users across several active subscriptions.

In region exploration, MIA-Explorer lets the user inspect a candidate region using the data the platform holds: OpenStreetMap geometries, census data and GTFS (General Transit Feed Specification) feeds. Service area definition is the only geometric input we require. The user draws the intended service area or uploads geodata.

Demand is not simulated endogenously but given as an input. The user can choose between uploading their own demand and using our prediction. The prediction is based on an origin-destination regression model (Zwick and Axhausen, 2022). The service area is divided into zones. Socioeconomic and accessibility attributes are computed for each of them, and expected trips are estimated for every origin-destination relation using a zero-inflated Poisson specification. The model is calibrated on over 1 million observed trips from MOIA's operational data from Hamburg. Because the resulting demand matrix is precalculated, the estimate updates live while the user changes the area.

In the simulation step (MIA-Simulator) the user is asked to configure parameters, each of which has a default value. How many to expose follows from who the user is. We build MIA for non-experts in simulation, who want to understand autonomous on-demand services and take initial planning decisions rather than conduct a detailed study. Detailed service planning remains in our consulting projects. That division is what allows us to expose the parameters in Table 1 and to fix everything else in the platform. After the user's input, everything downstream is automated. That process is described in the section "MIA simulation pipeline".

Table 1: The parameters MIA exposes to the user.

| Area           | Parameters exposed to the user                                                                                                                                                         |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Service design | Service area geometry, operating mode (ridehailing or ridepooling), stop mode (door-to-door or stop-based), distance between stops, maximum waiting time, maximum detour ratio (alpha) |
| Fleet          | Fleet size (including autoscaling towards a target service level), vehicle size, idle behaviour, rebalancing (on/off), battery capacity, battery discharge rate                        |
| Infrastructure | Hub locations, number of hubs and in-field chargers, ratio of AC to DC chargers                                                                                                        |
| Demand         | Demand level, temporal distribution of demand, demand scenario scaling                                                                                                                 |

The analysis step is supported by the MIA-Viewer (Figure 1), which generates for each simulation output a combined view of vehicle trajectories on a map, the key performance indicators (KPIs) and a set of charts. The

Commented [DZ1]: custom demand should be shown in the presentation. That's what the community probably cares even more about.

trajectories draw attention first, and they carry most of the interpretive weight: colour-coded occupancy and its changes let the user judge quickly whether a configuration is sensible. The KPIs include wait time, travel time, detour ratio, vehicle kilometres, trips served, acceptance rate and occupancy. Results feed an external costs calculator and a business case calculator, which combine them with local cost assumptions.



Figure 1: Screenshot of MIA-Viewer showing simulation output of a Berlin simulation run

## MIA simulation pipeline

The point of this section is the division of labour it implies. The entire scenario build, from network to demand matrix to simulation configuration, is automated, and the user contributes a geometry and a handful of fleet parameters. That is what makes the platform usable for the intended audience, and it is also what creates the problem the rest of this paper deals with: the parameters the user does not set are still set by someone, and their values still determine the result.

The pipeline starts once all parameters have been set by the user. It then generates the MATSim input files, starts the MATSim run on a server, and converts the simulation output into the format the viewer needs, including the configuration of the views.

First, the input files are generated. The road network is built from OpenStreetMap; the data is downloaded and filtered to include only the necessary links. For the stop network a grid with the defined distance between the lines is created, and at each crossing a stop is placed on the nearest link. The operation facilities file, which holds the hubs and their charging infrastructure, is created next, and the vehicles file is generated on that basis. The user never handles a file.

Then the files are uploaded to AWS and a MATSim run is started. If automatic fleet sizing and/or automatic hub placing is activated, the fleet and hubs are adjusted in every run. The automatic fleet sizing reacts to several KPIs and increases or decreases the fleet size accordingly. The same applies to the automatic hub placement.

## Guardrails against misinterpretation

Making the tool accessible has a flip side. In a few clicks, non-experts can produce results they are not equipped to interpret, and a KPI dashboard gives no indication of how far a result can be pushed. We use two layers to address this and plan to add two more.

The first layer is input guidance. The user can only set parameters between a lower and upper limit. The second is explanation inside the product rather than in a manual: every input field carries a description, and the methodological background is available in the application itself, where the decision is taken.

We are working on extending the guardrails by including warnings for parameter sets which do not fit together even though each individual parameter stays within its limit, e.g. high demand with a small fleet size, or a large service area with a low maximum waiting time. Another layer, which we would like to add, is automated interpretation of the results using large language models, so that the user receives the KPIs in context, together with a suggestion of what to adjust to make the service more efficient.

The latter, however, does not only mitigate the problem; it introduces a new form of it. An automatically generated interpretation reads as confident even when it is wrong, and a user who cannot check the KPIs is equally unable to check the text explaining them. Guardrails themselves require guardrails, and we regard this as an open question rather than a solved one.

## Outlook and discussion

The next development step is to move from fleet simulation with exogenous demand to a full mobility simulation with endogenous demand in the wider transport system. The reason is a user requirement rather than a technical one: modal shift (especially from public transport to on-demand services, or cannibalisation), intermodality and emissions can only be assessed if the rest of the transport system reacts to the service. This is planned work, and we report no results for it.

The synthetic population for this step is built on Volkswagen's Mobility Demand Creator, complemented by national travel survey data (e.g. Mobilität in Deutschland, MiD), and the approach has already been applied outside MIA in our consulting projects. Mode choice for DRT is calibrated using surveys we design ourselves and feed into the model.

This is where we would like input from the meeting, and we are asking about the system as it stands rather than about the roadmap. Three questions in particular. On the automated scenario build: what additional parameters would you expect? On presentation: which additional KPIs are missing from the dashboard, and are the current ones understandable? On guardrails: which parameter combinations would you warn about? We will bring a live demonstration to the meeting, and we would welcome anyone who wants to try to break it.

## References

Bischoff, J., Soeffker, N. and Maciejewski, M. (2016). A framework for agent based simulation of demand responsive transport systems. VSP Working Paper 2016-13, TU Berlin. <https://doi.org/10.14279/depositonce-5760>

Horni, A., Nagel, K. and Axhausen, K. W. (eds.) (2016). The Multi Agent Transport Simulation MATSim. Ubiquity Press, London.

Zwick, F. and Axhausen, K. W. (2022). Ride pooling demand prediction. Journal of Transport Geography.

Commented [DK2]: meinst du den MDC?

Commented [TK2R2]: hubs, yes

Commented [DZ3]: Könnte man rauslassen. Stattdessen: ..., which is already applied manually in our consulting projects.

Commented [DZ4]: seems like a very specific question.

Commented [DZ5]: I like the framing. It's an excellent call to action that does not sound too salesy, also for the presentation.